8.1 Defining and Using Sequences and Series

Name _____ Period ____ Date _____

Algebra II 8.1 · Student sheet

I can…

- write the terms of a sequence and write a rule for a sequence.
- write a series using summation notation.
- find the sum of a series.

Vocabulary · fill in the missing word

Sequence — an _____ list of numbers; each number is a term

Explicit rule — a _____ for the nth term, a_n, in terms of n

Series — the sum of the terms of a _____

Summation notation — Σ (i = 1 to n) a_i _____ $a_1 + a_2 + \ldots + a_n$; i is the index of summation

Part 1 · Toothpick squares

Build or draw a row of squares: 1 square, 2 squares, 3 squares, … Count the toothpicks.

Stage n	1	2	3	4	5	20
Toothpicks a_n						

Rule: a_n = _____ Total for stages 1–5: _____

Part 2 · Notes: fill in the blanks

A **sequence** is an ordered _____; a **series** is the _____ of its terms.

Σ (i = 1 to n) a_i means _____.

Σ (i = 1 to n) i = _____

Part 3 · Examples (work them with the class)

Example 1

Write the first six terms of each sequence.

a. $a_n = 2n + 5$ b. $f(n) = (-3)^{n-1}$

Example 2

Describe the pattern, write the next term, and write a rule for the nth term.

a. −5, 10, −15, 20, … b. 2, 6, 12, 20, …

Example 3

Write the series in summation notation: $4 + 8 + 12 + \ldots + 100$

Example 4

Find each sum.

a. $\Sigma\ (i = 1 \text{ to } 6)\ (i^2 + 1)$ b. $\Sigma\ (i = 1 \text{ to } 30)\ i$

Part 4 · Partner practice

Partner 1

Write a rule for the nth term: $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \ldots$

Partner 2

Find $\Sigma\ (k = 3 \text{ to } 7)\ (2k - 1)$.